



Data table components

Here's a **riddle** for you. What do a music playlist, a calendar agenda, and an email inbox have in common? I'll give you a hint. It's not a weekly jam session. The answer is they're all arranged in tables. Go ahead and check out your email inbox or a favorite playlist, or look at your calendar agenda. There's tables in every one! A data table, or tabular data, has a very simple structure. It's arranged in rows and columns. **You can call the rows "records" and the columns "fields."** They basically mean the same thing, but records and fields can be used for any kind of data table, while rows and columns are usually reserved for spreadsheets. When talking about structured databases, people in data analytics usually go with "records" and "fields." **Sometimes a field can also refer to a single piece of data, like the value in a cell.** In any case, you'll hear both versions of these terms used throughout this program and your job. Let's go back to our playlist example. We'll use the new terms we just introduced. So each song is a record. Each record has the same fields as the other records in the same order. In other words, the playlist has the same information about each song.

Each song characteristic, like the title and the artist, is a field. Each separate field has the same data type, but different fields can have different types. Let me show you what I mean. For the song list, the song titles are a text or string type, while the song's length

could be a number type if you're using it for calculations. Or it could be a date and time type. The column for favorites is Boolean since it has two possible values: favorite or not favorite. We can view spreadsheets in the same way. The records in a spreadsheet might be about all sorts of things: clients, products, invoices, or anything else. Each record has several fields, which reveal more about the clients, products, or invoices. The value in every cell contains a specific piece of data, like the address of a client or the dollar amount of an invoice. As a data analyst, lots of data will come your way, and records, fields, and values in data tables will help you navigate analysis. Understanding the structures of the tables you're working with is a part of that. And hopefully, while you're working hard on your analysis and those tables, you can have a little fun with a different data table: the one with your favorite playlist!